## CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** For m = 5, the Pythagorean triplet $(2m,\ m^2-1,\ m^2+1)$ is: 1

**(a)** 10, 24, 26 **(b)** 5, 24, 26 **(c)** 10, 26, 27 **(d)** 10, 23, 27

**2.** The sum of the cubes of the first 5 natural numbers is: 1

**(a)** 15 **(b)** 225 **(c)** 230 **(d)** 30

**3.** How many non-square numbers lie between $16^2$ and $17^2$? 1

**(a)** 32 **(b)** 16 **(c)** 33 **(d)** 31

**4.** The value of $25^2 - 24^2$ is: 1

**(a)** 25 **(b)** 50 **(c)** 49 **(d)** 48

**5.** The triangular number $T_{12}$ (sum of 1 to 12) is: 1

**(a)** 90 **(b)** 66 **(c)** 144 **(d)** 78

**6.** The sum of the consecutive triangular numbers $T_{11}$ and $T_{12}$ is: 1

**(a)** 145 **(b)** 121 **(c)** 144 **(d)** 156

**7.** For m = 6, the Pythagorean triplet $(2m,\ m^2-1,\ m^2+1)$ is: 1

**(a)** 6, 35, 37 **(b)** 12, 37, 38 **(c)** 12, 35, 37 **(d)** 12, 34, 38

**8.** The sum of the cubes of the first 6 natural numbers is: 1

**(a)** 441 **(b)** 42 **(c)** 21 **(d)** 447

**9.** How many non-square numbers lie between $20^2$ and $21^2$? 1

**(a)** 39 **(b)** 41 **(c)** 20 **(d)** 40

**10.** The value of $31^2 - 30^2$ is: 1

**(a)** 31 **(b)** 61 **(c)** 60 **(d)** 62

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the Pythagorean triplet whose smallest member is 16. 2

**12.** Express $4^3$ as the sum of 4 consecutive odd numbers. 2

**13.** Express $225$ as the sum of consecutive odd numbers starting from 1. 2

**14.** Find the Pythagorean triplet whose smallest member is 20. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Show that $T_{10} + T_{11}$ is a perfect square. 3

**16.** Check whether $(28, 195, 197)$ is a Pythagorean triplet. 3

**17.** Find $\sqrt{3136}$ by prime factorisation. 3

OR

Show that $T_{12} + T_{13}$ is a perfect square. 3

**18.** Check whether $(30, 224, 226)$ is a Pythagorean triplet. 3

OR

Find $\sqrt{4356}$ by prime factorisation. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Verify that $13832$ can be written as the sum of two cubes in two different ways: $2^3 + 24^3$ and $18^3 + 20^3$. 5

OR

Find $\sqrt[3]{21952}$ by prime factorisation and verify the result by cubing. 5

**20.** Verify that $20683$ can be written as the sum of two cubes in two different ways: $10^3 + 27^3$ and $19^3 + 24^3$. 5

OR

Find $\sqrt[3]{32768}$ by prime factorisation and verify the result by cubing. 5

## END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND